2) Temos que a a função de Bessel de primeira espécie é definida por:

$$J_p(x) = \sum_{m=0}^{+\infty} \frac{(-1)^m}{m!\,(m+p)!} \left(\frac{x}{2}\right)^{2m+p} \quad ; \; p = 0, 1, 2 \ldots$$

Para $p = 1$

$$\boxed{J_1(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,(m+1)!} \left(\frac{x}{2}\right)^{2m+1}}$$

• Propriedade de derivação:

$$\boxed{\frac{d}{dx}\left(J_p(x)\right) = J_{p-1}(x) - \frac{p}{x} J_p(x)}$$

$$\frac{d^2}{dx^2}\left(\frac{d\left(J_p(x)\right)}{dx}\right) = \frac{d\left(J_{p-1}(x)\right)}{dx} - \frac{d}{dx}\left(\frac{p}{x} J_p(x)\right) = J_{p-2}(x) - \left(\frac{(p-1)}{x} J_{p-1}(x)\right) - \left(-\frac{p}{x^2} J_p(x) + \frac{d J_p(x)}{dx} \frac{p}{x}\right)$$

$$\frac{d^2\left(J_p(x)\right)}{dx^2} = J_{p-2}(x) - \left(\frac{(p-1)}{x} J_{p-1}(x)\right) + \frac{p}{x^2} J_p(x) - \left(\frac{p}{x}\left(J_{p-1}(x) - \frac{p}{x} J_p(x)\right)\right)$$

$$= J_{p-2}(x) - \left(\frac{(p-1)}{x} J_{p-1}(x)\right) + \frac{p}{x^2} J_p(x) - \frac{p}{x} J_{p-1}(x) + \left(\frac{p}{x}\right)^2 J_p(x)$$

$$= \boxed{J_p(x)\left(\frac{p(1+p)}{x^2}\right) - J_{p-1}(x)\left(\frac{2p-1}{x}\right) + J_{p-2}(x)}$$

• Propriedade

$$\boxed{J_{-p}(x) = (-1)^p J_p(x)}$$

Para $p = 1$, temos que as derivadas são:

$$\boxed{\frac{d}{dx}\left(J_1(x)\right) = J_0(x) - \frac{1}{x} J_1(x)}$$

Onde:

$$\left\{\begin{array}{l} J_1(x) = \displaystyle\sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,(m+1)!} \left(\frac{x}{2}\right)^{2m+1} \\ J_0(x) = \displaystyle\sum_{m=0}^{+\infty} \frac{(-1)^m}{m!\;m!} \left(\frac{x}{2}\right)^{2m} \\ J_{-1}(x) = -J_1(x) \end{array}\right.$$

$$\boxed{\begin{aligned} \frac{d^2}{dx^2}\left(J_1(x)\right) &= \left(\frac{2}{x^2}\right) J_1(x) - \frac{1}{x} J_0(x) - J_1(x) \\ &= J_1(x)\left(\frac{2}{x^2} - 1\right) - \frac{1}{x} J_0(x) \end{aligned}}$$

Aproximação das derivadas por diferenças finitas

i) Aproximação central com 3 pontos:

$$\left.\frac{dJ_1(x)}{dx}\right|_{x=x_i} = \frac{J_1(i+1) - J_1(i-1)}{2\Delta x}$$

$$\left.\frac{d^2J_1(x)}{dx^2}\right|_{x=x_i} = \frac{J_1(i-1) - 2J_1(i) + J_1(i+1)}{\Delta x^2}$$

ii) Aproximação Forward com 3 pontos:

$$\frac{dJ_1(x)}{dx} = \frac{-3J_1(i) + 4J_1(i+1) - J_1(i+2)}{2\Delta x}$$

$$\frac{d^2J_1(x)}{dx^2} = \frac{J_1(i) - 2J_1(i+1) + J_1(i+2)}{\Delta x^2}$$

iii) Aproximação central com 5 pontos

Fazendo as expansões de Taylor para os pontos $i+2$ e $i-2$:

$$T(x+2\Delta x) = T(x) + \left.\frac{dT}{dx}\right|_x 2\Delta x + \left.\frac{d^2T}{dx^2}\right|_x \frac{(2\Delta x)^2}{2} + \ldots \quad (1)$$

$$T(x-2\Delta x) = T(x) - \left.\frac{dT}{dx}\right|_x 2\Delta x + \left.\frac{d^2T}{dx^2}\right|_x \frac{(2\Delta x)^2}{2} + \ldots \quad (2)$$

Fazendo $(1)-(2)$:

$$T(x+2\Delta x) - T(x-2\Delta x) = 2\left.\frac{dT}{dx}\right|_x \cdot 2\Delta x \Rightarrow \boxed{\frac{dT}{dx} = \frac{T(x+2\Delta x) - T(x-2\Delta x)}{4\Delta x}}$$

Fazendo $(1)+(2)$:

$$T(x+2\Delta x) + T(x-2\Delta x) = 2T(x) + 2\left.\frac{d^2T}{dx^2}\right|_x \frac{4\Delta x^2}{2} \Rightarrow \boxed{\left.\frac{d^2T}{dx^2}\right|_x = \frac{T(x+2\Delta x) + T(x-2\Delta x) - 2T(x)}{4\Delta x^2}}$$

iv) Aproximação backward com 3 pontos

Para o ponto $i-1$:

$$T(x-\Delta x) = T(x) - \left.\frac{dT}{dx}\right|_x \Delta x + \left.\frac{d^2T}{dx^2}\right|_x \frac{\Delta x^2}{2} + \ldots \quad (1)$$

$$T(x-2\Delta x) = T(x) - \left.\frac{dT}{dx}\right|_x 2\Delta x + \left.\frac{d^2T}{dx^2}\right|_x \frac{(2\Delta x)^2}{2} \quad (2)$$

O sistema resultante é:

$$\begin{bmatrix} -\Delta x & \frac{\Delta x^2}{2} \\ -2\Delta x & 2\Delta x^2 \end{bmatrix} \begin{Bmatrix} \left.\frac{dT}{dx}\right|_x \\ \left.\frac{d^2T}{dx^2}\right|_x \end{Bmatrix} = \begin{Bmatrix} T(x-\Delta x) - T(x) \\ T(x-2\Delta x) - T(x) \end{Bmatrix}$$

O determinante da matriz é:

$$det = -2\Delta x^3 - (-\Delta x^3) = -2\Delta x^3 + \Delta x^3$$
$$= -\Delta x^3$$

$$\begin{array}{ccc} & (2\times 2) & (2\times 1) \\ \begin{Bmatrix} \left.\frac{dT}{dx}\right|_x \\ \left.\frac{d^2T}{dx^2}\right|_x \end{Bmatrix} = -\frac{1}{\Delta x^3} & \begin{bmatrix} 2\Delta x^2 & -\frac{\Delta x^2}{2} \\ 2\Delta x & -\Delta x \end{bmatrix} & \begin{Bmatrix} T(x-\Delta x) - T(x) \\ T(x-2\Delta x) - T(x) \end{Bmatrix} \end{array}$$

$$\left.\frac{dT}{dx}\right|_x = -\frac{1}{\Delta x^3}\left[2\Delta x^2 (T(x-\Delta x) - T(x)) - \frac{\Delta x^2}{2}(T(x-2\Delta x) - T(x))\right] = -\frac{1}{\Delta x}\left(2T(x-\Delta x) - 2T(x) - \frac{T(x-2\Delta x)}{2} + \frac{T(x)}{2}\right)$$

$$= -\frac{1}{\Delta x}\left(\frac{4T(x-\Delta x) - 4T(x) - T(x-2\Delta x) + T(x)}{2}\right) = \boxed{\frac{-4T(x-\Delta x) + 3T(x) + T(x-2\Delta x)}{2\Delta x}}$$

$$\left.\frac{d^2T}{dx^2}\right|_x = -\frac{1}{\Delta x^3}\left[2\Delta x(T(x-\Delta x) - T(x)) - \Delta x(T(x-2\Delta x) - T(x))\right] = -\frac{1}{\Delta x^2}\left[2T(x-\Delta x) - 2T(x) - T(x-2\Delta x) + T(x)\right]$$

$$= \boxed{\frac{-2T(x-\Delta x) + T(x) + T(x-2\Delta x)}{\Delta x^2}}$$

Questão 4

→ Canto inferior

$$K\left(\frac{T^* - 2T_i + T_{i+1}}{\Delta x^2} + \frac{T^* - 2T_i + T_{i+mx}}{\Delta y^2}\right) + S = 0$$

Fluxo em x:

$$-K\left.\frac{dT}{dx}\right|_{x=i} = q(x_i) \Rightarrow -K\left(\frac{T_{i+1} - T^*}{2\Delta x}\right)(-1) = q$$

$$T_{i+1} - T^* = \frac{q \cdot 2\Delta x}{K}$$

$$\boxed{T^* = T_{i+1} - \frac{2q\Delta x}{K}}$$

Fluxo em y:

$$h[T(L)-T_\infty] = -k \left.\frac{dT}{dy}\right|_L$$

$$h[T_i - T_\infty](-1) = -k\left(\frac{T_{i+mx} - T^\circ}{2\Delta y}\right)$$

$$T_{i+mx} - T^\circ = \frac{2h\Delta y}{k}(T_i - T_\infty)$$

$$T^\circ = T_{i+mx} - \frac{2h\Delta y}{k}(T_i - T_\infty)$$

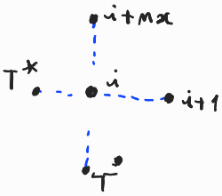
$$k\left(\frac{T^* - 2T_i + T_{i+1}}{\Delta x^2} + \frac{T^\circ - 2T_i + T_{i+mx}}{\Delta y^2}\right) + S = 0$$

$$\frac{1}{\Delta x^2}T^* - \frac{2}{\Delta x^2}T_i + \frac{1}{\Delta x^2}T_{i+1} + \frac{1}{\Delta y^2}T^\circ - \frac{2T_i}{\Delta y^2} + \frac{T_{i+mx}}{\Delta y^2} = \frac{-S}{K}$$

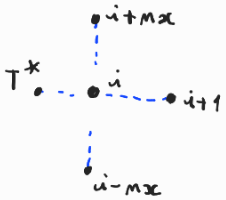
$$\frac{1}{\Delta x^2}\left(T_{i+1} - \frac{2q\Delta x}{k}\right) - \frac{2}{\Delta x^2}T_i + \frac{1}{\Delta x^2}T_{i+1} + \frac{1}{\Delta y^2}\left(T_{i+mx} - \frac{2h\Delta y}{K}T_i + \frac{2h\Delta y T_\infty}{K}\right) - \frac{2T_i}{\Delta y^2} + \frac{T_{i+mx}}{\Delta y^2} = -\frac{S}{K}$$

$$T_{i+1}\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta x^2}\right) + T_i\left(-\frac{2}{\Delta x^2} - \frac{2h}{\Delta y k} - \frac{2}{\Delta y^2}\right) + T_{i+mx}\left(\frac{1}{\Delta y^2} + \frac{1}{\Delta y^2}\right) = -\frac{S}{k} + \frac{2q}{\Delta x k} - \frac{2hT_\infty}{k\Delta y}$$

$$\begin{cases} A_{i,i+1} = -\dfrac{2}{\Delta x^2} \\[2ex] A_{i,i} = \dfrac{2}{\Delta x^2} + \dfrac{2h}{\Delta y k} + \dfrac{2}{\Delta y^2} \\[2ex] A_{i,(i+mx)} = -\dfrac{2}{\Delta y^2} \end{cases}$$



→ Fronteira esquerda (exceto os cantos)



$$k\left(\frac{T^* - 2T_i + T_{i+1}}{\Delta x^2} + \frac{T_{i-mx} - 2T_i + T_{i+mx}}{\Delta y^2}\right) + S = 0$$

Fluxo em x:

$$-K\left.\frac{dT}{dx}\right|_{x=i} = q(x_i) \Rightarrow -K\left(\frac{T_{i+1} - T^*}{2\Delta x}\right)(-1) = q$$

$$T_{i+1} - T^* = \frac{q \cdot 2\Delta x}{k}$$

$$\boxed{T^* = T_{i+1} - \frac{2q\Delta x}{k}}$$

$$\frac{1}{\Delta x^2}\left(T_{i+1} - \frac{2q}{\Delta x K}\right) - \frac{2T_i}{\Delta x^2} + \frac{T_{i+1}}{\Delta x^2} + \frac{T_{i-mx}}{\Delta y^2} - \frac{2T_i}{\Delta y^2} + \frac{T_{i+mx}}{\Delta y^2} = -\frac{S}{K}$$

$$T_{i+1}\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta x^2}\right) + T_i\left(-\frac{2}{\Delta x^2} - \frac{2}{\Delta y^2}\right) + T_{i+mx}\left(\frac{1}{\Delta y^2}\right) + T_{i-mx}\left(\frac{1}{\Delta y^2}\right) = -\frac{S}{k} + \frac{2q}{\Delta x k}$$

$$\begin{cases} A_{i,i+1} = -\dfrac{2}{\Delta x^2} \; ; \; A_{i,i-mx} = -\dfrac{1}{\Delta y^2} \\[2ex] A_{i,i} = \dfrac{2}{\Delta x^2} + \dfrac{2}{\Delta y^2} \\[2ex] A_{i,i+mx} = -\dfrac{1}{\Delta y^2} \end{cases}$$

→ Frontera derecha (hecho en central)

$$k\left(\frac{T_{i-1}-2T_i+T^*}{\Delta x^2}+\frac{T_{i-mx}-2T_i+T_{i+mx}}{\Delta y^2}\right)+S=0$$

Flujo en x:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dx}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^*-T_{i-1}}{2\Delta x}\right)$$

$$T^*=-\frac{2h\Delta x}{k}(T_i-T_\infty)+T_{i-1}$$

$$\frac{T_{i-1}}{\Delta x^2}-\frac{2T_i}{\Delta x^2}+\frac{1}{\Delta x^2}T_{i-1}-\frac{1}{\Delta x^2}\frac{2h\Delta x\,T_i}{k}+\frac{2h}{k\Delta x}T_\infty+\frac{T_{i-mx}}{\Delta y^2}-\frac{2T_i}{\Delta y^2}+\frac{T_{i+mx}}{\Delta y^2}=-\frac{S}{k}$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}+\frac{1}{\Delta x^2}\right)+T_i\left(\frac{-2}{\Delta x^2}-\frac{2h}{k\Delta x}-\frac{2}{\Delta y^2}\right)+T_{i-mx}\left(\frac{1}{\Delta y^2}\right)+T_{i+mx}\left(\frac{1}{\Delta y^2}\right)=-\frac{S}{k}-\frac{2hT_\infty}{\Delta x k}$$

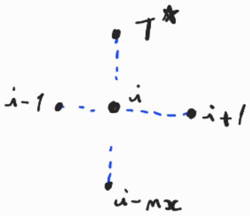
$$A_{i,i-1}=\left(\frac{-2}{\Delta x^2}\right)$$

$$A_{i,i}=\frac{2}{\Delta x^2}+\frac{2h}{k\Delta x}+\frac{2}{\Delta y^2}$$

$$A_{i,i-mx}=-\frac{1}{\Delta y^2}$$

$$A_{i,i+mx}=-\frac{1}{\Delta y^2}$$

→ Frontera superior



$$k\left(\frac{T_{i-1}-2T_i+T_{i+1}}{\Delta x^2}+\frac{T^*-2T_i+T_{i-mx}}{\Delta y^2}\right)+S=0$$

Flujo en y:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^*-T_{i-mx}}{2\Delta y}\right)$$

$$T^*=-\frac{2h\Delta y}{k}(T_i-T_\infty)+T_{i-mx}$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)-2T_i\left(\frac{1}{\Delta x^2}\right)+T_{i+1}\left(\frac{1}{\Delta x^2}\right)-\frac{2hT_i}{k\Delta y}+\frac{2hT_\infty}{k\Delta y}+\frac{T_{i-mx}}{\Delta y^2}-\frac{2T_i}{\Delta y^2}+\frac{T_{i-mx}}{\Delta y^2}=-\frac{S}{k}$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}-\frac{2h}{k\Delta y}-\frac{2}{\Delta y^2}\right)+T_{i+1}\left(\frac{1}{\Delta x^2}\right)+T_{i-mx}\left(\frac{1}{\Delta y^2}+\frac{1}{\Delta y^2}\right)=-\frac{S}{k}-\frac{2hT_\infty}{k\Delta y}$$

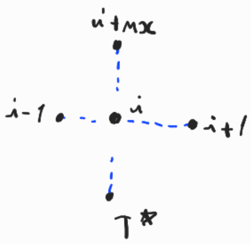
$$A_{i,i-1}=-\frac{1}{\Delta x^2}$$

$$A_{i,i}=\frac{2}{\Delta x^2}+\frac{2h}{k\Delta y}+\frac{2}{\Delta y^2}$$

$$A_{i,i+1}=-\frac{1}{\Delta x^2}$$

$$A_{i,i-mx}=-\frac{2}{\Delta y^2}$$

→ Frontera inferior



$$k\left(\frac{T_{i-1}-2T_i+T_{i+1}}{\Delta x^2}+\frac{T^*-2T_i+T_{i+mx}}{\Delta y^2}\right)+S=0$$

Flujo en y:

$$-h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$-h[T_i-T_\infty]=-k\left(\frac{T_{i+mx}-T^*}{2\Delta y}\right)$$

$$T^*=-\frac{2h\Delta y}{k}(T_i-T_\infty)+T_{i+mx}$$

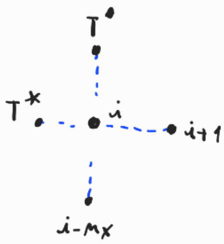
$$k\left(\frac{T_{i-1}-2T_i+T_{i+1}}{\Delta x^2}+\frac{T^*-2T_i+T_{i+mx}}{\Delta y^2}\right)+S=0$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}\right)+T_{i+1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(\frac{-2h}{\Delta y k}\right)+\frac{2hT_\infty}{k\Delta y}+\frac{T_{i+mx}}{\Delta y^2}-T_i\left(\frac{2}{\Delta y^2}\right)+T_{i+mx}\left(\frac{1}{\Delta y^2}\right)=-\frac{S}{k}$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}-\frac{2h}{\Delta y k}-\frac{2}{\Delta y^2}\right)+T_{i+1}\left(\frac{1}{\Delta x^2}\right)+T_{i+mx}\left(\frac{2}{\Delta y^2}\right)=-\frac{S}{k}-\frac{2hT_\infty}{k\Delta y}$$

$$\begin{cases} A_{i,i-1}=-\dfrac{1}{\Delta x^2} \\ A_{i,i}=\dfrac{2}{\Delta x^2}+\dfrac{2h}{\Delta y K}+\dfrac{2}{\Delta y} \\ A_{i,i+1}=-\dfrac{1}{\Delta x^2} \\ A_{i,i+mx}=-\dfrac{2}{\Delta y^2} \end{cases}$$

→ Cantó superior esquerre



$$k\left(\frac{T^*-2T_i+T_{i+1}}{\Delta x^2}+\frac{T_{i-mx}-2T_i+T^\circ}{\Delta y^2}\right)+S=0$$

Flux en x:

$$k\left(\frac{T_{i+1}-T^*}{2\Delta x}\right)=q \Rightarrow T^*=-\frac{2\Delta x q}{k}+T_{i+1}$$

Em y: Flux en y:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^\circ-T_{i-mx}}{2\Delta y}\right)$$

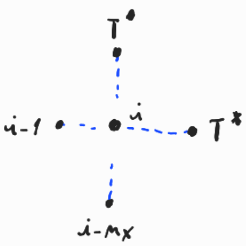
$$T^\circ=-\frac{2h\Delta y}{K}(T_i-T_\infty)+T_{i-mx}$$

$$-\frac{2q}{\Delta x}+\frac{T_{i+1}}{\Delta x^2}-\frac{2T_i}{\Delta x^2}+\frac{T_{i+1}}{\Delta x^2}+\frac{T_{i-mx}}{\Delta y^2}-\frac{2T_i}{\Delta y^2}-\frac{2hT_i}{\Delta y k}+\frac{2hT_\infty}{\Delta y k}+\frac{T_{i-mx}}{\Delta y^2}=-\frac{S}{k}$$

$$T_{i+1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(\frac{-2}{\Delta x^2}-\frac{2}{\Delta y^2}-\frac{2h}{\Delta y k}\right)+T_{i+1}\left(\frac{1}{\Delta x^2}\right)+T_{i-mx}\left(\frac{1}{\Delta y^2}+\frac{1}{\Delta y^2}\right)=-\frac{S}{k}-\frac{2hT_\infty}{\Delta y k}+\frac{2q}{\Delta x}$$

$$\begin{cases} A_{i,i+1}=-\dfrac{2}{\Delta x^2} \\ A_{i,i}=\dfrac{2}{\Delta x^2}+\dfrac{2}{\Delta y^2}+\dfrac{2h}{\Delta y K} \\ A_{i,i-mx}=-\dfrac{2}{\Delta y^2} \end{cases}$$

→ Cantó superior dret



$$k\left(\frac{T_{i-1}-2T_i+T^*}{\Delta x^2}+\frac{T_{i-mx}-2T_i+T^\circ}{\Delta y^2}\right)+S=0$$

Flux en x:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^*-T_{i-1}}{2\Delta x}\right)$$

$$T^*=-\frac{2h\Delta x}{K}(T_i-T_\infty)+T_{i-1}$$

Flux en y:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^\circ-T_{i-mx}}{2\Delta y}\right)$$

$$T^\circ=-\frac{2h\Delta y}{K}(T_i-T_\infty)+T_{i-mx}$$

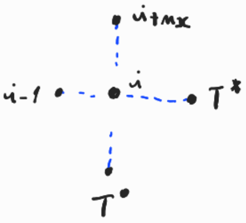
$$K\left(\frac{T_{i-1}-2T_i+T^*}{\Delta x^2}+\frac{T_{i-mx}-2T_i+T^\circ}{\Delta y^2}\right)+S=0$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}\right)+T_i\left(\frac{-2h}{\Delta x k}\right)+\frac{2hT_\infty}{\Delta x k}+T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_{i-mx}\left(\frac{1}{\Delta y^2}\right)+T_i\left(\frac{-2}{\Delta y^2}\right)+T_i\left(\frac{-2}{K\Delta y}\right)+\frac{2hT_\infty}{K\Delta y}+\frac{T_{i-mx}}{\Delta y^2}=-\frac{S}{K}$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}+\frac{1}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}-\frac{2h}{\Delta x K}-\frac{2}{\Delta y^2}-\frac{2h}{k\Delta y}\right)+T_{i-mx}\left(\frac{1}{\Delta y^2}+\frac{1}{\Delta y^2}\right)=-\frac{S}{K}-\frac{2hT_\infty}{\Delta x K}-\frac{2hT_\infty}{k\Delta y}$$

$$\begin{cases} A_{i,i-1}=-\dfrac{2}{\Delta x^2} \\ A_{i,i}=-\dfrac{2}{\Delta x^2}-\dfrac{2h}{\Delta x k}-\dfrac{2}{\Delta y^2}-\dfrac{2h}{k\Delta y} \\ A_{i,i-mx}=-\dfrac{2}{\Delta y^2} \end{cases}$$

→ Canto inferior direito



$$K\left(\frac{T_{i-1}-2T_i+T^*}{\Delta x^2}+\frac{T^\circ-2T_i+T^\circ}{\Delta y^2}\right)+S=0$$

Fluxo em y:

$$-h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=k\left(\frac{T_{i+mx}-T^\circ}{2\Delta y}\right)$$

$$T^\circ=\frac{-2h\Delta y}{K}(T_i-T_\infty)+T_{i+mx}$$

Fluxo em x:

$$h[T(L)-T_\infty]=-k\left.\frac{dT}{dy}\right|_L$$

$$h[T_i-T_\infty]=-k\left(\frac{T^*-T_{i-1}}{2\Delta x}\right)$$

$$T^*=\frac{-2h\Delta x}{K}(T_i-T_\infty)+T_{i-1}$$

$$K\left(\frac{T_{i-1}-2T_i+T^*}{\Delta x^2}+\frac{T^\circ-2T_i+T_{i+mx}}{\Delta y^2}\right)+S=0$$

$$T_{i-1}\left(\frac{1}{\Delta x^2}\right)+T_i\left(\frac{-2}{\Delta x^2}\right)-\frac{2hT_i}{k\Delta x}+\frac{2hT_\infty}{k\Delta x}+\frac{T_{i-1}}{\Delta x^2}-\frac{2hT_i}{\Delta y k}+\frac{2hT_\infty}{k\Delta y}+T_i\left(\frac{-2}{\Delta y^2}\right)+\frac{T_{i+mx}}{\Delta y^2}+\frac{T_{i+mx}}{\Delta y^2}=-\frac{S}{K}$$

$$T_{i-1}\left(\frac{2}{\Delta x^2}\right)+T_i\left(-\frac{2}{\Delta x^2}-\frac{2h}{k\Delta x}-\frac{2h}{\Delta y k}-\frac{2}{\Delta y^2}\right)+T_{i+mx}\left(\frac{2}{\Delta y^2}\right)=-\frac{S}{K}-\frac{2hT_\infty}{k\Delta x}-\frac{2hT_\infty}{k\Delta y}$$

$$\begin{cases} A_{i,i-1}=-\dfrac{2}{\Delta x^2} \\ A_{i,i}=\dfrac{2}{\Delta x^2}+\dfrac{2h}{k\Delta x}+\dfrac{2h}{\Delta y k}+\dfrac{2}{\Delta y^2} \\ A_{i,i+mx}=-\dfrac{2}{\Delta y^2} \end{cases}$$

Questão 5

i) Abordagem explícita

Neste caso conhecemos a temperatura em um instante k (T_i^k) e não conhecemos no instante $k+1$.

Para essa abordagem, as aproximações das derivadas são:

$$\frac{\partial T}{\partial t} = \frac{T_i^{k+1} - T_i^k}{\Delta t} \quad ; \quad \frac{\partial^2 T}{\partial x^2} = \frac{T_{i-1}^k - 2T_i^k + T_{i+1}}{\Delta x^2}$$

$$\lambda = \frac{k}{\rho c} \quad ; \quad \beta = \frac{\lambda \Delta t}{\Delta x^2}$$

Logo a temperatura no ponto i, para o instante $k+1$ é:

$$T_i^{k+1} = T_i^k + \beta\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right] + \frac{S\Delta t}{\rho c}$$

Condições de contorno:

$$\frac{\partial T}{\partial t} - \lambda \frac{\partial^2 T}{\partial x^2} = 0$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=0,t} = q(t) \quad ; \quad q(t) = \begin{cases} 1, & t \leq 10 \\ 0, & t > 10 \end{cases}$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=1,t} = 0 \quad ; \quad T(x, t=0) = \bar{T}$$

Utilizando uma discretização de 3 pontos:

Para o ponto da fronteira esquerda em $t=k$

$$T_i^{k+1} = T_i^k + \beta\left[T^{*k} - 2T_i^k + T_{i+1}^k\right]$$

Da condição de contorno, temos que fixado k:

$$\frac{\partial T}{\partial x} = \frac{T_{i+1}^k - T^{*k}}{2\Delta x} = -\bar{q} \Rightarrow T^{*k} = T_{i+1}^k + \bar{q} \cdot 2\Delta x$$

$$\boxed{T_i^{k+1} = T_i^k + \beta\left[T_{i+1}^k + \frac{\bar{q}}{2\Delta x} - 2T_i^k + T_{i+1}^k\right]}$$

Para o ponto central i em $t=k$:

$$\boxed{T_i^{k+1} = T_i^k + \beta\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right]}$$

Para o ponto da fronteira direita em $t=k$

$$\boxed{T_i^{k+1} = T_i^k + \beta\left[T_{i-1}^k - 2T_i^k + T^{*k}\right] = T_i^k + 2\beta\left[T_{i-1}^k - T_i^k\right]}$$

$$\left.\frac{\partial T}{\partial x}\right|_{x=1,t} = 0 \Rightarrow \frac{T^* - T_{i-1}}{2\Delta x} = 0 \Rightarrow T^* = T_{i-1}$$

$$\beta = \alpha \frac{\Delta t}{\Delta x^2} \leq 0{,}5$$

$$\frac{\Delta t}{\Delta x^2} \leq \frac{0{,}5}{0{,}01} = 50$$

$$\Delta x^2 \geq \frac{1}{50} = 0{,}02$$

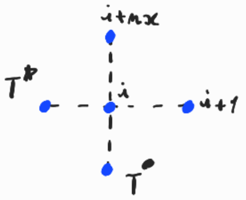
$$\Delta x \geq 0{,}14$$

$$\frac{L}{N} \geq 0{,}14 \Rightarrow N \leq \frac{1}{0{,}14} = 7{,}07$$

↳ Questão 6

Para a abordagem explícita Canto inferior esquerdo:

$$\frac{\partial T}{\partial t} = \frac{T_i^{k+1} - T_i^k}{\Delta t}$$



Em x:

$$h\left[T_i^k - T_\infty^k\right](-1) = -k\left(\frac{T_{i+1}^k - T^{*k}}{2\Delta x}\right)$$

$$T_{i+1}^k - T^{*k} = \frac{2\Delta x}{k} h\left[T_i^k - T_\infty^k\right]$$

$$T^{*k} = T_{i+1}^k - \frac{2\Delta x}{k} h\left[T_i^k - T_\infty^k\right]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T^{*k} - 2T_i^k + T_{i+1}^k\right]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T_{i+1}^k - \frac{2\Delta x h T_i^k}{k} + \frac{2\Delta x h T_\infty^k}{k} - 2T_i^k + T_{i+1}^k\right] \quad (1)$$

Em y:

$$h\left[T_i^k - T_\infty^k\right] = k\left(\frac{T_{i+mx}^k - T^\circ}{2\Delta y}\right)$$

$$T^\circ = T_{i+mx}^k - \frac{2\Delta y}{k} h\left[T_i^k - T_\infty^k\right]$$

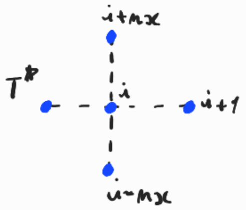
$$T_i^{k+1} = T_i^k + \beta y\left[T^\circ - 2T_i^k + T_{i+mx}^k\right]$$

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i+mx}^k - \frac{2\Delta y h T_i^k}{k} + \frac{2\Delta y h T_\infty^k}{k} - 2T_i^k + T_{i+mx}^k\right] \quad (2)$$

Somando (1) e (2):

$$\boxed{T_i^{k+1} = \left[T_i^k + \beta x\left[T_{i+1}^k - \frac{\Delta x h T_i^k}{k} + \frac{\Delta x h T_\infty^k}{k} - T_i^k\right] + \beta y\left[T_{i+mx}^k - \frac{\Delta y h T_i^k}{k} + \frac{\Delta y h T_\infty^k}{k} - T_i^k\right]\right]}$$

Borda esquerda:



Em x:

$$h\left[T_i^k - T_\infty^k\right](-1) = -k\left(\frac{T_{i+1}^k - T^{*k}}{2\Delta x}\right)$$

$$T_{i+1}^k - T^{*k} = \frac{2\Delta x}{k} h\left[T_i^k - T_\infty^k\right]$$

$$T^{*k} = T_{i+1}^k - \frac{2\Delta x}{k} h\left[T_i^k - T_\infty^k\right]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T^{*k} - 2T_i^k + T_{i+1}^k\right]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T_{i+1}^k - \frac{2\Delta x h T_i^k}{k} + \frac{2\Delta x h T_\infty^k}{k} - 2T_i^k + T_{i+1}^k\right] \quad (1)$$

Em y:

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-mx}^k - 2T_i^k + T_{i+1}^k\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = \frac{1}{2}\left[2T_i^k + 2\beta x\left(T_{i+1}^k - \frac{\Delta x h T_i^k}{k} + \frac{\Delta x h T_\infty^k}{k} - T_i^k\right) + \beta y\left[T_{i-mx}^k - 2T_i^k + T_{i+1}^k\right]\right]}$$

Canto superior esquerdo

Em x:

$$h[T_i^k - T_\infty^k](-1) = -k\left(\frac{T_{i+1}^k - T^{*k}}{2\Delta x}\right)$$

$$T_{i+1}^k - T^{*k} = \frac{2\Delta x}{k} h[T_i^k - T_\infty^k]$$

$$T^{*k} = T_{i+1}^k - \frac{2\Delta x}{k} h[T_i^k - T_\infty^k]$$

$$T_i^{k+1} = T_i^k + \beta x[T^{*k} - 2T_i^k + T_{i+1}^k]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T_{i+1}^k - \frac{2\Delta x h T_i^k}{k} + \frac{2\Delta x h T_\infty^k}{k} - 2T_i^k + T_{i+1}^k\right] \quad (1)$$

Em y:

$$h[T_i^k - T_\infty^k] = k\left(\frac{T^{\circ k} - T_{i-mx}^k}{2\Delta y}\right)$$

$$T^{\circ k} = T_{i-mx}^k + \frac{2\Delta y h}{k}[T_i^k - T_\infty^k]$$

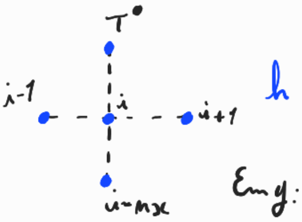
$$T_i^{k+1} = T_i^k + \beta y[T_{i-nx}^k - 2T_i^k + T^{\circ k}]$$

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-mx}^k - 2T_i^k + T_{i-mx}^k + \frac{2\Delta y h T_i^k}{k} - \frac{2\Delta y h T_\infty^k}{k}\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = T_i^k + \beta x\left[T_{i+1}^k - \frac{\Delta x h T_i^k}{k} + \frac{\Delta x h T_\infty^k}{k} - T_i^k\right] + \beta y\left[T_{i-mx}^k - T_i^k + \frac{\Delta y h T_i^k}{k} - \frac{\Delta y h T_\infty^k}{k}\right]}$$

Borda superior



Em x:

$$T_i^{k+1} = T_i^k + \beta x[T_{i-1}^k - 2T_i^k + T_{i+1}^k]$$

Em y:

$$h[T_i^k - T_\infty^k] = k\left(\frac{T^{\circ k} - T_{i-mx}^k}{2\Delta y}\right)$$

$$T^{\circ k} = T_{i-mx}^k + \frac{2\Delta y h}{k}[T_i^k - T_\infty^k]$$

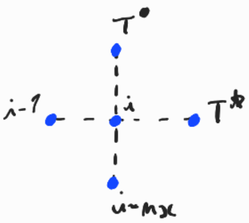
$$T_i^{k+1} = T_i^k + \beta y[T_{i-nx}^k - 2T_i^k + T^{\circ k}]$$

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-mx}^k - 2T_i^k + T_{i-mx}^k + \frac{2\Delta y h T_i^k}{k} - \frac{2\Delta y h T_\infty^k}{k}\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = \frac{1}{2}\left(2T_i^k + \beta x[T_{i-1}^k - 2T_i^k + T_{i+1}^k] + 2\beta y\left[T_{i-mx}^k - T_i^k + \frac{\Delta y h T_i^k}{k} - \frac{\Delta y h T_\infty^k}{k}\right]\right)}$$

Canto superior direito



Em x:

$$h[T_i^k - T_\infty^k](-1) = -k\left(\frac{T^{*k} - T_{i-1}^k}{2\Delta x}\right)$$

$$T^{*k} - T_{i-1}^k = \frac{2\Delta x}{k} h[T_i^k - T_\infty^k]$$

$$T^{*k} = T_{i-1}^k + \frac{2\Delta x}{k} h[T_i^k - T_\infty^k]$$

$$T_i^{k+1} = T_i^k + \beta x[T_{i-1}^k - 2T_i^k + T^{*k}]$$

$$T_i^{k+1} = T_i^k + 2\beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x h T_i^k}{k} - \frac{\Delta x h T_\infty^k}{k}\right] \quad (1)$$

Em y: $h[T_i^k - T_\infty^k] = k\left(\frac{T^{\circ k} - T_{i-mx}^k}{2\Delta y}\right)$

$$T^{\circ k} = T_{i-mx}^k + \frac{2\Delta y\, h\,[T_i^k - T_\infty^k]}{k}$$

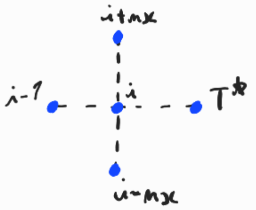
$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-nx}^k - 2T_i^k + T^{\circ k}\right]$$

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-mx}^k - 2T_i^k + T_{i-mx}^k + \frac{2\Delta y\, h\, T_i^k}{k} - \frac{2\Delta y\, h\, T_\infty^k}{k}\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = T_i^k + \beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x\, h\, T_i^k}{k} - \frac{\Delta x\, h\, T_\infty^k}{k}\right] + \beta y\left[T_{i-mx}^k - T_i^k + \frac{\Delta y\, h\, T_i^k}{k} - \frac{\Delta y\, h\, T_\infty^k}{k}\right]}$$

Borda direita



Em x: $h[T_i^k - T_\infty^k](-1) = -k\left(\frac{T^{*k} - T_{i-1}^k}{2\Delta x}\right)$

$$T^{*k} - T_{i-1}^k = \frac{2\Delta x}{k}\, h\,[T_i^k - T_\infty^k]$$

$$T^{*k} = T_{i-1}^k + \frac{2\Delta x}{k}\, h\,[T_i^k - T_\infty^k]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T_{i-1}^k - 2T_i^k + T^{*k}\right]$$

$$T_i^{k+1} = T_i^k + 2\beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x\, h\, T_i^k}{k} - \frac{\Delta x\, h\, T_\infty^k}{k}\right] \quad (1)$$

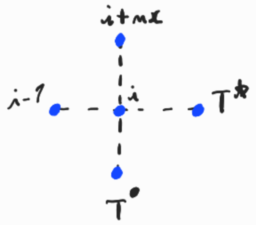
Em y:

$$T_i^{k+1} = T_i^k + \beta y\left[T_{i-nx}^k - 2T_i^k + T_{i+mx}\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = \frac{1}{2}\left(2T_i^k + 2\beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x\, h\, T_i^k}{k} - \frac{\Delta x\, h\, T_\infty^k}{k}\right] + \beta y\left[T_{i-nx}^k - 2T_i^k + T_{i+mx}\right]\right)}$$

Canto inferior direito



Em x: $h[T_i^k - T_\infty^k](-1) = -k\left(\frac{T^{*k} - T_{i-1}^k}{2\Delta x}\right)$

$$T^{*k} - T_{i-1}^k = \frac{2\Delta x}{k}\, h\,[T_i^k - T_\infty^k]$$

$$T^{*k} = T_{i-1}^k + \frac{2\Delta x}{k}\, h\,[T_i^k - T_\infty^k]$$

$$T_i^{k+1} = T_i^k + \beta x\left[T_{i-1}^k - 2T_i^k + T^{*k}\right]$$

$$T_i^{k+1} = T_i^k + 2\beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x\, h\, T_i^k}{k} - \frac{\Delta x\, h\, T_\infty^k}{k}\right]$$

Em y: $h[T_i^k - T_\infty^k] = k\left(\frac{T_{i+mx}^k - T^{\circ}}{2\Delta y}\right)$

$$T^{\circ k} = T_{i+mx}^k - \frac{2\Delta y\, h\,[T_i^k - T_\infty^k]}{k}$$

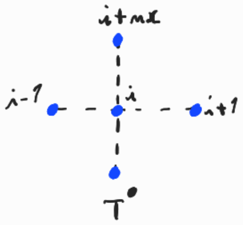
$$T_i^{k+1} = T_i^k + \beta y\left[T^{\circ k} - 2T_i^k + T_{i+mx}^k\right]$$

$$T_i^{k+1} = T_i^k + 2\beta y\left[T_{i+mx}^k - \frac{\Delta y\, h\, T_i^k}{k} + \frac{\Delta y\, h\, T_\infty^k}{k} - T_i^k\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = T_i^k + \beta x\left[T_{i-1}^k - T_i^k + \frac{\Delta x\, h\, T_i^k}{k} - \frac{\Delta x\, h\, T_\infty^k}{k}\right] + \beta y\left[T_{i+mx}^k - \frac{\Delta y\, h\, T_i^k}{k} + \frac{\Delta y\, h\, T_\infty^k}{k} - T_i^k\right]}$$

Bordo inferior



Em x: $T_i^{k+1} = T_i^k + \beta_x\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right]$

Em y: $h\left[T_i^k - T_\infty^k\right] = k\left(\frac{T_{i+mx}^k - T^{\bullet}}{2\Delta y}\right)$

$$T^{\bullet k} = T_{i+mx}^k - \frac{2\Delta y\, h}{k}\left[T_i^k - T_\infty^k\right]$$

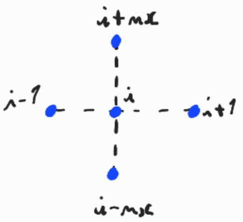
$$T_i^{k+1} = T_i^k + \beta_y\left[T^{\bullet k} - 2T_i^k + T_{i+mx}^k\right]$$

$$T_i^{k+1} = T_i^k + 2\beta_y\left[T_{i+mx}^k - \frac{\Delta y\, h\, T_i^k}{k} + \frac{\Delta y\, h\, T_\infty^k}{k} - T_i^k\right]$$

Somando as equações:

$$\boxed{T_i^{k+1} = 2T_i^k + \beta_x\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right] + 2\beta_y\left[T_{i+mx}^k - \frac{\Delta y\, h\, T_i^k}{k} + \frac{\Delta y\, h\, T_\infty^k}{k} - T_i^k\right]}$$

Pontos internos



Em x: $T_i^{k+1} = T_i^k + \beta_x\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right]$

Em y: $T_i^{k+1} = T_i^k + \beta_y\left[T_{i-mx}^k - 2T_i^k + T_{i+mx}^k\right]$

Somando as equações:

$$\boxed{T_i^{k+1} = \frac{1}{2}\left[2T_i^k + \beta_x\left[T_{i-1}^k - 2T_i^k + T_{i+1}^k\right] + \beta_y\left[T_{i-mx}^k - 2T_i^k + T_{i+mx}^k\right]\right]}$$